

C2-02 Advanced SQL - Start Here

Begin here before ASQL1

WHY THIS MODULE EXISTS

Course 1 SQL taught you to retrieve and summarize data. C2-02 teaches the senior control layer: predict grain, prevent join fan-out, structure complex logic, use windows safely, investigate data quality, validate independently and package SQL another analyst can inherit.

EXACT ROUTE

ASQL1 book -> matching SQL practice task -> save genuine attempt + validation -> explain-back -> protected review -> close review -> changed retry. Repeat through ASQL12. Then two cumulative Mini-Labs -> Gate A. B/C remain protected fresh reassessments.

DIALECT BOUNDARY

Use PostgreSQL 18/current syntax. The preserved Alex video is MySQL-oriented. Do not copy MySQL-specific syntax into the PostgreSQL lab when the book/reference differs. The video is never a hidden prerequisite.

C2-02 prerequisites and tools

Advanced does not mean unexplained

YOU SHOULD ALREADY KNOW

SELECT, WHERE, GROUP BY, HAVING, aggregates, basic INNER/LEFT JOIN and simple subqueries. You should be able to read a small SQL result and describe what one row represents.

YOU NEED

PostgreSQL 18 preferred, plus a SQL client. The next build step will supply schema/setup SQL, independent practice SQL workspace, Mini-Labs, protected reviews and Gate data.

RECOVERY

If fundamentals are weak, do not memorize these advanced examples. Repair the specific Course 1 skill first. If PostgreSQL is unavailable, mark runtime blocked; static reading does not count as hands-on SQL mastery.

C2-02 mastery / Gate contract

No query-run checkmark and no percentage gaming

LESSON PASS

Runnable query + stated input/output grain + independent validation query + one plausible failure mode + 60-90 second explain-back + changed retry. A query that merely executes is Developing.

GATE A

Independent integrated analytical SQL on fresh schema/data. You must prove grain, joins, time/KPI logic, quality and validation.

B/C RETEST

Fresh domains and values. Targeted repair never reveals the next full Gate automatically. One unresolved critical grain/denominator/validation failure blocks pass.